

Chapter 10

Writing a Thesis

The highlight of mathematical writing at university is the report on a final year project, called a **thesis** or **dissertation**. This document has a distinctive structure, in between a short book and a research paper. A thesis surveys a body of advanced literature and presents original work. At undergraduate level, usually the former outweighs the latter.

Writing a thesis is an irreplaceable experience in university education. This document can reveal a great deal about the author's knowledge, understanding, curiosity, and enthusiasm. Examination scripts do not convey so much information.

One doesn't learn how to write a thesis by reading an instruction manual. The thesis will take shape gradually, as a result of regular interactions between student and supervisor. The document's high-level organisation—number of chapters, outline of their content—is normally considered at a relatively early stage of the project. Planning will also suggest appropriate headings and will aid the sequencing of the arguments. Periodic reviews of work in progress may alter priorities, or even re-direct the research. The writing process culminates with the exercise of proof-reading, in itself a valuable, if painstaking, experience.

In this chapter I give an overview of the structure of a thesis, and then discuss selected aspects of writing—choosing the title, writing the abstract, compiling the bibliography—where analysing examples may be helpful. I also include a brief introduction to L^AT_EX, the software adopted by the mathematical publishing industry. Most theses are now typeset in L^AT_EX.

10.1 Theses and Other Publications

A thesis is a substantial document. Its length varies considerably, depending on level, topic, and institutional requirements. An undergraduate thesis could be some fifty pages long. There are also theses at master and doctoral level, called MSc and PhD theses respectively.¹

¹ Acronyms for Master of Science and Philosophy Doctor.

A good undergraduate thesis may rise to the level of a master thesis. A doctoral dissertation is longer, has greater depth, and more original research. The brief guidelines given in this chapter apply to any thesis. For further reading, see Higham's book [19], which devotes two chapters to PhD theses and has an extended bibliography.

A thesis is subdivided into heading, body, and closing matters. The heading supplies essential information: title, abstract, table of contents, acknowledgements. One may also find lists of mathematical symbols, figures, and tables, which some institutions require explicitly. Title and abstract define the subject matter; we'll consider them in some detail in Sects. 10.2 and 10.3. L^AT_EX will take care of the menial task of compiling the table of contents. The acknowledgements is a paragraph where the author thanks people for their guidance and support (the supervisor, typically), for useful discussions, for pointing out things, or for allowing the author to use their text in the thesis.

The body contains the bulk of the material, organised into chapters, sections, and, if appropriate, sub-sections. The first chapter is the introduction, written so as to make the thesis self-contained. Here we find motivations, background material and key references, enough details to be able to grasp the main results, and a description of the content of the rest of the thesis.

The length of individual chapters may vary, but each chapter must be a substantial component of the whole, with a clear identity. A brief closing chapter is desirable if not compulsory. This is an opportunity to re-visit ideas and results from an informed perspective, discuss limitations of the work, identify open problems, and chart directions of future research.

Any accessory material which should not interfere with the main text (data tables, computer programs, tedious proofs, etc.) is confined to appendices. The last item in a thesis is the bibliography, listing the sources cited in the text; it'll be considered in Sect. 10.4.

There is a symbiotic relationship between a thesis and a research paper; the higher the level of the thesis, the stronger the bond. At doctoral level, research papers lead to a thesis, or are extracted from it after the thesis is completed. At undergraduate level this connection is necessarily more remote, yet a research element must be present in every respectable thesis.

But a thesis is not a research paper. A research paper communicates new results to specialists, invariably within tight space constraints. It is seldom self-contained: sketchy motivations, unexplained jargon, and terse citations are commonplace. A thesis and a paper also differ in format. A paper is shorter, divided into sections, not chapters, while its simpler navigation renders a table of contents unnecessary.

A thesis is not a book either. A book covers thoroughly a body of established material. It is longer than a thesis, and we expect it to be wholly self-contained. A book begins with a preface, which is neither an abstract nor an introduction, but a bit of both. It is rare to find a book without exercises in each chapter, and even rarer one without a subject index at the end. (Notably, the celebrated book [15] has neither.) A textbook, being conceived as teaching material, may offer solutions or hints to exercises. This is a modern trend: hardly any of the classic, timeless mathematics textbooks provided solutions or hints to exercises—a fact worth pondering on. Some

advanced publications resemble theses: the *review articles*, published in dedicated research journals, and the *research monographs*, published as books, usually within a collection. These are a sort of advanced theses, which collect together previously published works. They usually appear when a new research area has reached maturity.

Most scholarly publications are specialised; they are not intended for the general public. So how narrow should the target audience of a thesis be? This is largely a question of personal taste. Conscious decisions must be made from the very beginning, because the title, the abstract, and the first paragraphs of the introduction will set the tone for the rest of the document.

Let us inspect the opening sentences of some theses and research papers, to see how strongly the character of the work is established with just few words. Later on, we shall analyse titles and abstracts from the same perspective.

Our first example is a master thesis (from [35], with minor editing).

Let $GL(d, q)$ denote the general linear group of invertible $d \times d$ -matrices over the finite field $\mathbb{F}_q$ with q elements. In computational group theory, it is of interest to calculate the order n of an element $A \in GL(d, q)$, i.e.,

$$n = \min\{j \in \mathbb{N} : A^j = I\}$$

where I is the $d \times d$ identity matrix. The integer d can be large, e.g., $d > 100$.

The style is formal, with definitions and symbols appearing straight away. The jargon (general linear group, finite field, order of a matrix, etc.) and the corresponding notation ($GL(d, q)$, $\mathbb{F}_q$, etc.) belong to standard introductory algebra. The author still provides brief explanations, which convey an impression of helpful, considerate writing. This document promises to be accessible to any mathematically mature reader, a desirable feature of any undergraduate thesis.

Our next example, from a thesis in geometry [16], illustrates a completely different expository strategy.

In 1984, Schechtman et al. [SGBC84] announced that the symmetry group of an aluminium-manganese alloy crystal, produced by rapid cooling, was that of the icosahedron. Such a symmetry is not possible for a periodic structure in three dimensions. This discovery brought down a long-held assumption in crystallography, that the only structures with some sense of long-range order were periodic.

The thesis begins with an account of the surprising outcome of a physics experiment. (We shall learn about literature citation in Sect. 10.4.) The author tells a story, skilfully building some drama. The near complete absence of mathematical terms—formal definitions and symbols are given much later—makes this excerpt very inviting. An engaging, accessible style is welcome in any publication, at any level.

A thesis can be highly specialised. One could use the established dictionary of a particular research area, without explanations or apologies, from the very beginning. This choice undoubtedly simplifies the writer's task, but it creates problems

for the reader, thereby reducing the readership basis. This is legitimate, as long as the jargon is used wisely to sharpen the exposition, not gratuitously to impress the reader.

The following opening passage is taken from a PhD thesis in the area of *ergodic theory* [3].

Ergodic optimisation is a branch of the ergodic theory of topological dynamics which is concerned with the study of T -invariant probability measures, and ergodic averages of real-valued functions f defined on the phase space of a dynamical systems $T : X \rightarrow X$, where X is a compact metric space.

This sentence explains the meaning of the expression ‘ergodic optimisation’, which features in the thesis’ title. To achieve this within a limited space, the author relies on the reader’s familiarity with half a dozen advanced concepts. Three important symbols are also defined in the same sentence. Undeniably, this thesis is for specialists.

In research papers, extremes of specialisation are accepted, or at least tolerated. Some editors insist—wisely, in my view—that the author provide clear motivations and a minimum of background, but this requirement is by no means universal. Papers that begin with a dry, unmotivated list of definitions are not uncommon. The following opening paragraph of standard definitions appears unchanged in three related publications [28–30].

Let K be an algebraic extension of $\mathbb{Q}_p$, and let $\mathcal{O}$ be its integer ring with maximal ideal $\mathcal{M}$, and residue field k . If $\bar{K}$ is an algebraic closure of K , we denote by $\bar{\mathcal{O}}$ and $\bar{\mathcal{M}}$ the integral closure of $\mathcal{O}$ in $\bar{K}$ and the maximal ideal of $\bar{\mathcal{O}}$, respectively.

In the following extreme example, the author disposes of definitions altogether [7].

1. **Introduction.** The notation used is that of [2]; in particular ...

One cannot proceed without the cited publication (by the same author). Only a committed reader will accept such a blunt treatment.

10.2 Title

The title provides an essential description of the content of a document. The titles of research papers compete for attention with other titles in journals, bibliographical lists, databases, etc. They help the reader decide whether to read on. A thesis has a captive readership—the examiners—so this problem does not arise immediately. Nonetheless, a good title can do a lot to give the thesis its unique identity, so it’s worth spending some time thinking about it.

A title is a short phrase without symbols. To get an idea of the wide range of possibilities, we now examine titles of various types of publications, and assess the suitability of the title for a thesis, possibly after amendments. Alongside each title we display the bibliographical data—see Sect. 10.4.

Introductory real analysis [A.N. Kolmogorov and S.V. Fomin, Dover Publications, New York (1970).] The term ‘introductory’ immediately calls to mind a textbook, so this term is out of place in a thesis’ title. The shorter title *Real analysis* would be much too general for a thesis, as it defines a vast area of mathematics.

Making transcendence transparent. (Subtitle: *An intuitive approach to classical transcendental number theory.*) [E. Burger and R. Tubbs, Springer-Verlag, New York (2004).] Subtitles can be quite effective. In this example, the title has an appealing alliteration, while the subtitle spells out the title’s promise. However, the title’s pedagogical slant and broad take are again the signature of a textbook, not a thesis.

Algebraic aspects of cryptography [N. Koblitz, Springer-Verlag, Berlin (1998).] The subject area covered by this textbook appears to be narrower than in the previous examples, whereas in fact it is still rather broad. If this title were used in a thesis, an expert would probably expect a literature survey. A thesis with an original research component would require a more specific title.

The arithmetic-geometric mean of Gauss. [T. Gilmore, MSc thesis, Queen Mary, University of London (2012).] As the reference to Gauss suggests, this topic is classical, and an expert will know that the arithmetic-geometric mean is connected to many deep mathematical theories. So this is a good title for a thesis, even if there is no indication of what might be the original contribution.

On the zeros of polynomials with restricted coefficients [P. Borwein and T. Erdelyi, *Illinois J. Math.* **41** (1997) 667–675.] This is a rare instance of a research problem with a concise and non-technical description, which makes a good title. The preposition ‘on’ lends an authoritative tone to it, promising some general result.

Toward a theory of Pollard’s ρ -method [Eric Bach, *J. Information and Computation* **90** (1991) 139–155.] A very eloquent title. It suggests that the method being considered was not supported by a theory, and that some progress has been made. This would be an ideal title for a thesis. The use of the symbol ρ is justified, because the aforementioned method is invariably associated with it.

A complete determination of the complex quadratic fields of class number one [H.M. Stark, *Michigan Math. J.* **14** (1967) 1–27.] The forceful, unambiguous title of this well-known publication announces the solution of a long-standing problem. Such an assertive style suits well a research paper with a strong result; the trimmed version *Quadratic fields of class number one* could be the title of a thesis.

Diophantine integrability [R.G. Halburd, *J. Phys A: Math. Gen.* **38** (2005) 1–7.] This title is short and effective. Each term has an established meaning in a specific area of mathematics, and the unexpected juxtaposition of the two terms suggests interdisciplinarity. This title would be perfect for a thesis.

Outer billiards on kites [R.E. Schwartz, volume 171 of *Annals of Mathematics Studies*, Princeton University Press, Princeton and Oxford (2009).] The title of this research monograph cleverly exploits double meanings. Both terms ‘Outer billiards’ and ‘kites’ are mathematical, but not many readers will be familiar with them. The unlikely significance that this phrase would have in ordinary English is bound to raise interest among a general mathematical audience.

Euclidean algorithms are Gaussian [V. Baladi and B. Vallée, *Journal of Number Theory* **110** (2005) 331–386.] Summarising a result with a sentence makes an

effective title for a research article, but perhaps less so for a thesis. The variant *Probabilistic aspects of Euclidean algorithms* is not so specific, and would work better for a thesis.

Metric number theory: the good and the bad. [R.E. Thorn, PhD thesis, Queen Mary, University of London (2004).] An excellent thesis title, enlivened by a colon. After announcing the research area (metric number theory), a clever reference to an epic Western provides a hint to the specific topic (badly approximable numbers).

To persist or not to persist? [J. Holzbauer and S.J. Schreiber, *Nonlinearity* **17** (2004) 1393–1406.] This paper is about ‘uniformly persistent vector fields’, and the author has transformed the subject matter into a Shakespearean question. Such a self-conscious title must be supported by strong content.

It is easy to determine whether a given integer is prime, [Andrew Granville *Bulletin of the American Math. Soc.* **42** (2005) 3–38.] Another title-sentence, with the added bonus of being provocative, because primality testing is a notoriously difficult problem. A fair deal of self-confidence is needed to choose a title of this kind. (This paper won the author the Chauvenet prize, awarded by the Mathematical Association of America to the author of an outstanding expository article on a mathematical topic.)

10.3 Abstract

An abstract is a short summary of a document. If well-written, it complements the title and sustains the reader’s attention. The abstract of a research paper is placed immediately after the title, on the front page. In a thesis, where space constraints are not so severe, the abstract usually appears on a separate page.

It is difficult to compose a convincing abstract before the work has been completed, understood, and written out. So the abstract—the first thing one reads—is typically the last thing to be written. The main results must be in evidence, expressed with precision and clarity; at the same time, specialised jargon should be reduced to a minimum, so as not to alienate potential readers. A delicate balance must be achieved between these conflicting requirements.

Abstracts are short, rarely exceeding 200 words. The use of symbols is to be reduced to a minimum, the space is confined and every word counts. The exercise of writing MICRO-ESSAYS (see Sect. 6.6) is good training for writing abstracts, as both tasks require similar rigour and discipline.

We now examine in detail abstracts taken from mathematical literature. They are less than optimal, and we seek to improve them. (The original text has been edited, mainly to protect anonymity.) We don’t intend to produce templates of well-written abstracts—this would be futile, as no two abstracts are the same. Rather, we want to sharpen our analytical skills, learn how to spot problems, and find ways of solving them. For this purpose, understanding the mathematics is not essential. Indeed, dealing with some unknown words and symbols has a certain advantage, as one must pay attention to the writing’s internal structure. We have met a

similar situation in Chap. 7, when we learnt how to organise the beginning of a proof.

Our first example, from a research paper, exemplifies a rather common problem: the use of unnecessary symbols.

ABSTRACT. Let F be a rational map of degree $n \geq 2$ of the Riemann sphere $\overline{\mathbb{C}}$. We develop a theory of equilibrium states for the class of Hölder continuous functions f for which the pressure is larger than $\sup f$. We show that there exists a unique conformal measure (reference measure) and a unique equilibrium state, which is equivalent to the conformal measure with a positive continuous density.

The fault is plain: the symbols F , n , and $\overline{\mathbb{C}}$ are introduced, but not used again in the abstract, breaking the first golden rule of symbol usage—see Sect. 6.2. Unused symbols distract rather than help. Even if these symbols are used again in the main text, the abstract is not the place where the notation is to be established, and these definitions should be deferred. On this account, the symbol f in the second sentence is used appropriately. Embedded in the second sentence we find the abbreviation ‘sup’, for ‘supremum’. This abbreviation (much like ‘max’, for maximum, or ‘lim’, for limit) is meant for formulae, and should not be mixed with words. So we replace it by the full expression.

Re-writing the first sentence without symbols and amending the second sentence do not require any understanding of the mathematics. We leave the third sentence as it is.

ABSTRACT. We consider rational maps of the Riemann sphere, of degree greater than 1. We develop a theory of equilibrium states for the class of Hölder continuous functions f for which the pressure is larger than the supremum of f . We show that ...

The symbol f in the second sentence is not strictly necessary, and can easily be disposed of.

We develop a theory of equilibrium states for the class of Hölder continuous functions for which the pressure is larger than the supremum of the function.

The next abstract belongs to a thesis. The author presents an accessible problem using limited jargon.

ABSTRACT. The logistic map is a well-studied map of the unit interval into itself. However, if we treat x as a discrete variable, as is done in any computer, then every orbit is eventually periodic. Thus the aperiodic behaviour that the continuous map displays for some value of the parameter r cannot be obtained from computer simulation. We investigated the differences and the similarities between the dynamics of a continuous map and its discrete approximation. We found that the limit cycles of a discrete map follow the unstable periodic orbits of the corresponding continuous map.

There are unknown words (logistic map, orbit, limit cycle), but also sufficiently many familiar terms (map, unit interval, eventually periodic, etc.) to help us discern the subject matter. The first sentence says that this work is about the ‘logistic map’, presented as a well-known object of investigation. The second and third sentences motivate the study. The author introduces the symbol x , but we are not told what it is. We guess that it must represent the argument of the logistic map, namely a point in the unit interval. Furthermore, this symbol is not used again in the abstract, so it is unnecessary. The symbol r suffers from similar problems. It’s a parameter, but we are not told which quantity depends on it. We guess that this is the logistic map, which is referred to as the ‘continuous map’, presumably meaning that its argument assumes a continuum, rather than a discrete set, of values. The symbol r is not used again either.

With two important ideas not in sharp focus, the main message of the thesis—if we discretise the domain of the logistic map, then we obtain new phenomena worth studying—gets a bit lost.

The closing sentence, which summarises the main findings, is vague. What does ‘we found’ mean? Presumably this is not a proof, otherwise the author would have stated that clearly. What does ‘follow’ mean? Presumably it indicates some form of convergence. Even if in the abstract there isn’t enough room for a complete statement of the results, we must find a way to render these vague statements acceptable. Finally, the author employs the past tense (we investigated, we found). In an abstract, the present tense is more common. Based on the above considerations, we rewrite the abstract as follows.

ABSTRACT. We consider the logistic map, a well-studied map of the unit interval which depends on a parameter. If the domain of this map is discretised, as happens in any computer simulation, then, necessarily, all orbits become eventually periodic. Thus the aperiodic orbits observed for certain parameter values no longer exist. We investigate differences and similarities between the original map and its discrete approximations. We provide evidence that the limit cycles of the discrete map converge to the unstable orbits of the original map, in a sense to be made precise.

The last expression states that the thesis contains experimental data or heuristic arguments that support a clearly formulated notion of convergence. (We hope that this is the case in the present thesis!)

Next we consider the opening sentences of an abstract of a research paper. The problems are more subtle here: there are no superfluous symbols, and we can’t rely on an understanding of the mathematics!

ABSTRACT. In this work we investigate properties of minimal solutions of multi-dimensional discrete periodic variational problems. A well-known one-dimensional example of such a problem is the Frenkel-Korontova model. We select a family ...

The opening sentence begins with the expression ‘in this work’; this is redundant and should simply be deleted. Expressions of this type serve a purpose only if the present

work is being juxtaposed to other works: ‘In 1964, Milnor proved an estimate...In this work, we prove...’

The object of the investigation is described by a long string of attributes: ‘multi-dimensional discrete periodic variational problem’. Such a flat arrangement of words requires a pause or a highlight. At this stage however, it’s not clear how this should be done, so we read on.

The second sentence begins with an expression of the type ‘an example of this is that’. We have already criticised this format in Sect. 1.3. As written, the emphasis is placed on the word ‘example’, but the terms ‘one-dimensional’ or ‘Frenkel-Korontova model’ surely are more significant. The author’s intentions are now clearer: the term ‘multi-dimensional’ in the first sentence is contrasted with ‘one-dimensional’ in the second sentence, and a known example is given for clarification. To get across the message that this work generalises ‘one-dimensional’ to ‘multi-dimensional’, we shall isolate the latter in the first sentence and emphasise the former in the second.

ABSTRACT. We investigate properties of minimal solutions of discrete periodic variational problems, in the multi-dimensional case. These generalise one-dimensional problems, such as the well-known Frenkel-Korontova model. We select a family ...

In our final abstract some symbols are necessary.

ABSTRACT. Let $f : X \rightarrow X$, $X = [0, 1)$, be an interval exchange transformation (IET) ergodic with respect to the Lebesgue measure on X . Let $f_t : X_t \rightarrow X_t$ be the IET obtained by inducing f to $X_t = [0, t)$, $0 < t < 1$. We show that

$$X_{wm} = \{0 < t < 1 : f_t \text{ is weakly mixing}\}$$

is a residual subset of X of full Lebesgue measure. The result is proved by establishing a Diophantine sufficient condition on t for f_t to be weakly mixing.

The overall content of the abstract is unclear, although we recognise several words introduced in Chaps. 2 and 4. These fragments of understanding are sufficient to clarify the logic of the argument.

The set X is the unit interval, and we have a function f of this set into itself. This function belongs to a certain class of functions, identified by the acronym IET. The author assumes that f has certain properties, specified by some jargon. Then a parameter t is introduced, and the same construction is repeated for a one-parameter family of functions f_t over sub-intervals X_t . The functions f_t , constructed from f via a process we need not worry about, are also of type IET. (Evidently the author believes that this assertion does not require justification.)

Using a Zermelo definition, the author introduces a subset X_{wm} of the open unit interval, determined by a property of f_t , called ‘weak mixing’. The subscript wm in the symbol X_{wm} must be an abbreviation for this expression. The paper establishes some properties of the set X_{wm} . An indication of the argument used in the proof is then given.

Our first task is to remove some symbols, including the displayed equation. This isn't too difficult; for example, the symbol X_{wm} is never used. The first two sentences are formal definitions, which need to be made more readable. The acronym IET is useful, since the long expression it represents is used twice. However, the primary object is 'interval exchange transformation', not IET, so the latter should follow the former, not precede it. Finally, in the expression 'Diophantine sufficient condition' the two adjectives compete with each other, so we separate them out.

ABSTRACT. Let f be an interval exchange transformation (IET) of the unit interval, ergodic with respect to the Lebesgue measure, and let f_t be the IET obtained by inducing f on the sub-interval $[0, t)$, with $0 < t < 1$. We show that the set of values of t for which f_t is weakly mixing is a residual subset of full Lebesgue measure. The result is proved by establishing a Diophantine condition on t , which is sufficient for weak mixing.

More examples of abstracts in need of improvement are given as exercises.

10.4 Citations and Bibliography

Scholarly writing invariably requires reference to other sources. This is carried out in two stages. First, the source is mentioned in the text in an abbreviated manner. Second, a full ordered list of all cited sources (books, journal papers, web pages, etc.) is presented at the end of the document, in a section headed 'Bibliography' or 'References'.

There are some conventions for structuring citations and for writing the bibliography. We begin with a passage involving references to books.

We briefly review some properties of p -adic numbers, introduced by Hensel in 1908 [18]. Background material may be found in Gouvea's (2000) [14]; the arithmetical side of the theory is developed in Hasse (1980) [17, Chap. 7].

Each item enclosed in square brackets is a citation, which refers to a bibliographical entry. Citations should not intrude upon the text, and a sentence in which all citations have been removed should still read well. The citations [14] and [18] refer to books as a whole, whereas [17] refers to a specific chapter in a book. The targeted item within a book may also be a theorem [15, theorem 191], a page [15, p. 156], a range of pages ([15, pp. 156–158], [15, 156ff]²), etc. The bibliographical data corresponding to the text above are organised as follows:

[14] F.Q. Gouvêa, *p-adic Numbers*, Springer-Verlag, Berlin (2000).

⋮

[17] H. Hasse, *Number Theory*, Springer-Verlag, Berlin (1980).

² The suffix *ff* to a page number means 'and the following pages'.

[18] K. Hensel, *Theorie der Algebraischen Zhalen*, Teubner, Leipzig and Berlin (1908).

The entries are usually ordered alphabetically by the author's surname. Each item contains the following information: label, author(s), title, publisher, publisher's home town, year of publication. In our case the labels are numbers (there are other possibilities, see below), and the book's title is typeset in italic font. In [17] the publisher is 'Springer-Verlag', located in Berlin. As the publishing industry becomes global, the publisher's address is becoming increasingly irrelevant and publishers with multiple addresses are commonplace. The book identifier International Standard Book Number (ISBN), often given, is helpful for searches in libraries, catalogues, and databases.

There are other ways to cite books, as in the following shorter variant of our paragraph.

We briefly review some properties of p -adic numbers, introduced in [18]. Background material may be found in [14]: the arithmetical side of the theory is developed in [17, Chap. 7].

To simplify the citations we have removed the author's name and the year of publication from the text. However, reading has become more laborious because the reference number conveys no information about the source. The following labelling method offers a compromise, providing essential information in abbreviated form.

We briefly review some properties of p -adic numbers, introduced in [Hen08]. Background material may be found in [Gou80]; the arithmetical side of the theory is developed in [Has80, Chap. 7].

Citations to journal papers follow the same rules given for books, although the bibliographical information is slightly different. Here is one example:

[32] P. Morton and J.H. Silverman, Periodic points, multiplicities, and dynamical units, *J. Reine Angew. Math.*, **461** (1995) 81–122.

The data to be supplied comprises: label, author, title of paper, name of journal, volume number, year of publication, page numbers. The title is in Roman font, whereas the name of the journal appears in slanted font (a variant of italics), usually in a standard abbreviated form. Thus *J. Reine Angew. Math.* stands for *Journal für die reine und angewandte Mathematik*.³ The volume number is in boldface. Page numbers are normally given in full, but the style 361–7 (rather than 361–367) is allowed. As for books, there is an identifier for journals, called International Standard Serial Number (ISSN).

Other sources for citations are articles in electronic journals [10], articles in conference proceedings [33], books in collections [4], reprints of books [9], theses [3], etc. There are great variations in the style of a bibliography. The best thing is to look around for examples of bibliographical entries of the various types by different authors and publishers.

³ This is the oldest mathematics journal still in existence, founded in 1826.

For unpublished documents on the web there is as yet no agreed format. The main information is the Uniform Resource Locator (URL), which identifies a web page uniquely, as in the following example:

[27] Text processing using L^AT_EX, Dept. of Engineering, University of Cambridge, <http://www.eng.cam.ac.uk/help/tpl/textprocessing/> (January 2014).

This item is a collection of resources for electronic typesetting (see Sect. 10.5). The title is followed by the name of the institution providing the service and the URL, typeset in ‘typewriter’ font. There is no reference to authors—too many to be listed here. A date is provided. Web pages don’t always carry dates (this one does), and even if they do, there is no way of deciding whether the displayed date corresponds to the last revision. The best one can do in this situation is to state when the web source was last accessed.

We present variants of bibliographical entries for web resources:

[8] P. Cvitanović, R. Artuso, R. Mainieri, G. Tanner, G. Vattay, *Chaos: Classical and Quantum*, chaosbook.org (Niels Bohr Institute, Copenhagen 2012).

[21] W. Jung, Core entropy and biaccessibility of quadratic polynomials, preprint (2014), [arXiv:1401.4792v1](https://arxiv.org/abs/1401.4792v1), 46 pages.

Item [8] is a web-book, and the citation format adheres to the explicit instructions provided on the site. Instead of the URL there is a *domain name* registered in the Domain Name System (DNS), which provides a more intuitive identification of a web resource. Reference [8] has a large number of authors; to mention the authors in the text, an abbreviation is necessary: ‘In Cvitanović et al. [8]....’⁴ The web-book [8] has been available for several years, and it is gradually evolving; this document structure would be impossible within the confines of traditional publishing.

The item [21] is an unpublished paper. It is stored in a large, open-access archive of preprints of scientific papers, called [arXiv](https://arxiv.org/), supported by Cornell University Library. Each preprint in this archive has a unique 8-digit identifier, plus a version number (v1, in this example) which reflects the fact that papers may be updated. Most preprints in the [arXiv](https://arxiv.org/) will eventually become available in a more permanent form. So, when proof-reading bibliographical entries, one should check that the papers in the [arXiv](https://arxiv.org/) haven’t been published. Some editors accept only citations to permanent and accessible material.

A word of warning. When quoting from the web, be aware that there is a lot of rubbish there, and most web pages have no guarantee of correctness. Also keep in mind that anyone looking for basic information on a subject (e.g., *abelian groups*) can—without any assistance—type the relevant heading in a search engine and access the corresponding entry in Wikipedia. For this reason, citations to a main on-line encyclopaedia tend to be superfluous, and one should consider more substantial sources.

⁴ Abbreviation for the Latin *et alii*, meaning ‘and others’.

10.4.1 Avoiding Plagiarism

There is no copyright in mathematical ideas, the law of intellectual property being primarily commercial. But there is copyright in using other people's words, pictures, and software. The act of publishing material produced by others as if it were your own is a serious academic offence, called **plagiarism**.

To respect copyright when using other people's words one must name the author, cite the source, and make clear which are the quoted words (either by quotation marks or by display). For extensive use of someone else's text (more than a paragraph, say), one must obtain the explicit permission of the copyright holder, and then insert a suitable remark in the acknowledgements section: 'I thank XXX for permission to quote YYY'.

The instant availability of free material on the web has increased considerably the risk of plagiarism caused by carelessness and naïvety rather than dishonesty. When cutting and pasting from other people's text, it is essential to keep a record of the source. A text stored without a note of its source could easily be used as our own text at a later time. As articles, books, and theses are now submitted in electronic form, it is possible to detect this type of plagiarism using specialised software.

10.5 T_EX and L^AT_EX

The programming language T_EX, developed by D. E. Knuth [23], and its user-friendly offspring L^AT_EX, written by L. Lamport [25], are the standard software in mathematical and scientific publishing. This book was typeset in L^AT_EX.

This typesetting system is intended for text containing mathematical formulae. It also supports the high-level features needed to manage a complex document: generating tables of contents, numbering of chapters and equations, cross referencing, handling bibliographical data, etc.

Let us inspect a very short L^AT_EX source file⁵:

```
\documentclass[12pt]{article}
\begin{document}
In 1735, Leonhard Euler proved the following
remarkable formula:
\begin{equation}
\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4}
+ \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}.
\end{equation}
\end{document}
```

⁵ The source file for the *Mathematical Writing* book contains over half a million characters.

The information is stored in plain text. We must type it in a file, and then run the L^AT_EX program, which generates a new file suitable for display, typically in PDF format.⁶ The end result is the following text, placed at the top of an empty page with the page number 1 at the bottom:

In 1735, Leonhard Euler proved the following remarkable formula:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}. \quad (1)$$

The contrast between the awkwardness of the L^AT_EX source and the neatness of the output may be off-putting at first. On closer acquaintance, it will instead become clear why this logical and powerful language has become so widespread.

In the source file we note frequent occurrences of the distinctive backslash symbol ‘\’, followed by a word. These are L^AT_EX commands, which instruct the program to do something. We see that, after an initial preamble, a pair of commands delimit the beginning and the end of the document. Within the document, another pair delimits a displayed equation, and within the equation various commands generate mathematical symbols and structures. Certain things occur automatically, such as the centering and numbering of the displayed equation, and the output is insensitive to the layout of the source code.

Of course, the software does a lot more than typesetting. Thus L^AT_EX may be expanded by incorporating auxiliary *libraries* of commands for specific tasks, such as B^IB^TE_X, which manages bibliographical information. Similarly, the appearance of a document may be altered by means of *macros*. These are libraries transparent to the user, which reset the layout parameters (font type and size, line and paragraph spacing, style of bibliographical list, etc.). Publishers use this facility to convert standard L^AT_EX source files into documents with the desired appearance.

Learning L^AT_EX takes some time, but if you plan to write a thesis this is a worthwhile investment. There is no shortage of introductory material: the Cambridge University L^AT_EX page is a good place to start [27].

Exercise 10.1 The following abstracts are poorly written. Identify the problems, hence write an improved version.

1. In this project the author examines some properties of continued fractions (CF). In the beginning, definitions, notations, and basic results and theorems are shown. Periodic continued fractions and best approximations are examined subsequently in depth. We examined a number of applications to mathematics and astronomy.
2. The most representative example of a 2-dimensional area-preserving twist map is the standard map, which is studied. Orbits for which the momentum p grows linearly (plus a periodic function) are shown to exist, classified and determined numerically. These orbits are the accelerator modes. The linear stability of these orbits is determined. The range in the parameter values for which they exist is also determined.

⁶ This is an established format for document representation.

3. We consider equal parameter generalized quadrangles, $GQ(s, s)$. All GQ s of order 2 and 3, $GQ(2, 2)$ and $GQ(3, 3)$, are known. It is conjectured all the GQ of order five are known as well. The known GQ of order 5 is the symplectic GQ , $W(5)$. $W(5)$ along with its dual are conjectured to be the only GQ s of order 5. The construction of a symplectic GQ is given and then used to construct the known GQ of order 5. Information about GQ was gathered, including some basic combinatorics, affine GQ , and incidence matrices in an attempt to prove the above conjecture.
4. Under the boundary condition on the initial value $T_k(0)$ ($T_k(0) > 0$; $k = 1, 2, \dots$) that $T_k(0) \rightarrow 0$ ($k \rightarrow \infty$), we integrate the semi-infinite system of nonlinear differential equations $\dot{T}_k = 2T_k(T_{k+1} - T_{k-1})$, ($k = 1, 2, \dots$; $T_0 = 0$) to obtain their general solution. We further investigate the asymptotic time behaviours of this general solution as $t \rightarrow \pm\infty$.

Exercise 10.2 This is the longest opening sentence of a mathematical abstract I managed to find in the literature [39].

By Morava's point of view on the stable homotopy category, the quotient in some sense associated to the filtration related to the height of formal group laws is studied by the category of modules over the function ring of the deformation space of the Honda group law of height n with the lift of the action of the automorphism group on the closed fibre through the Adams-Novikov spectral sequence.

1. Can you break it into shorter sentences?
2. Can you find a longer opening sentence in any mathematical publication?

Exercise 10.3 Explore the statement (4.31), and write a one-page report on your findings.

Exercise 10.4 This exercise proposes some MICRO-PROJECTS in experimental mathematics. You are given a sequence to study; your task is to generate numerically a sufficient number of terms, analyse the data, and then write a succinct report—three/four pages, say.

In the report you must define the problem, give a concise account of your findings, and synthesise them in a precise mathematical conjecture. Invariably, some plots will be necessary. You should provide at least one bibliographical item, and include the numerical code in an appendix.

For an introduction to experimental mathematics, see [42]; the book [5] is a good source of computational problems suitable for projects.

1. EULER'S POLYNOMIAL. Let $p(n) = n^2 + n + 41$ be Euler's polynomial (see Sect. 7.6), and let a_n be the number of prime values assumed by p over the first n natural numbers, that is,

$$a_n = \#\{k \in \mathbb{N} : k \leq n, \ p(k) \text{ is prime}\} \quad n \geq 1.$$

Refine conjecture 3 of Sect. 7.6 by identifying non-trivial bounds of the form

$$f(n) \leq a_n \leq g(n) \quad (10.1)$$

to hold for all sufficiently large n . The functions f and g must be chosen so as to minimise the growth rate of the difference $g - f$.

2. EXPONENTIAL SUMS. Given a real sequence (a_k) we define a sequence (z_n) of complex numbers via the sum

$$z_n = \sum_{k=1}^n e^{2\pi i a_k} \quad n \geq 1.$$

The number z_n may be interpreted as the end-point of a walk on the complex plane, consisting of n steps of unit length. Study (z_n) for one of the following sequences (a_k) :

- a. $a_k = k^2/50$
- b. $a_k = k^2/(50 + 1/100)$
- c. $a_k = k \log(k)$.

Formulate a conjecture concerning the behaviour of z_n or $|z_n|$ as n tends to infinity.

3. PRIMES AND POWERS OF TEN. Let a_n be as in (9.4).

- a. Prove that $f(\mathbb{N}) \subset A$, where

$$A = \mathbb{N} \setminus (2\mathbb{N} \cup (2 + 3\mathbb{N}) \cup 5\mathbb{N}). \quad (10.2)$$

- b. Provide experimental evidence that $f(\mathbb{N})$ is unbounded.
- c. Provide experimental evidence that $f(\mathbb{N}) = A$.